AI and IoT Solutions for Foundries & Casting | FoundryCast AI

FoundryCast AI delivers AI and IoT software for foundries and casting operations with RFID workforce identification, furnace access control, mold and tooling tracking, alloy and scrap inventory management, casting work-in-progress visibility, heat lot traceability, BLE, LoRaWAN, GPS yard tracking, edge computing, and MES/ERP integration.

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AI and IoT Workforce Identification, Asset Tracking, and Heat Lot Traceability for Foundries & Casting Operations

Foundries represent one of the most demanding manufacturing environments within the Primary Metals Industry. Every production cycle involves continuous movement of raw materials, charge materials, molds, flasks, core boxes, ladles, crucibles, production tooling, work-in-progress castings, finished components, forklifts, and personnel across multiple manufacturing areas. Maintaining accurate identification throughout these operations is essential for production efficiency, workforce safety, product quality, regulatory compliance, and customer documentation.

Typical production begins with receiving pig iron, steel scrap, return scrap, aluminum ingots, ferroalloys, and alloying additions before progressing through charge preparation, induction or cupola furnace melting, melt chemistry adjustment, slag removal, inoculation or degassing where applicable, mold preparation, pouring, cooling, shakeout, fettling, shot blasting, heat treatment, machining, dimensional inspection, non-destructive testing (NDT), warehouse storage, and customer shipment.

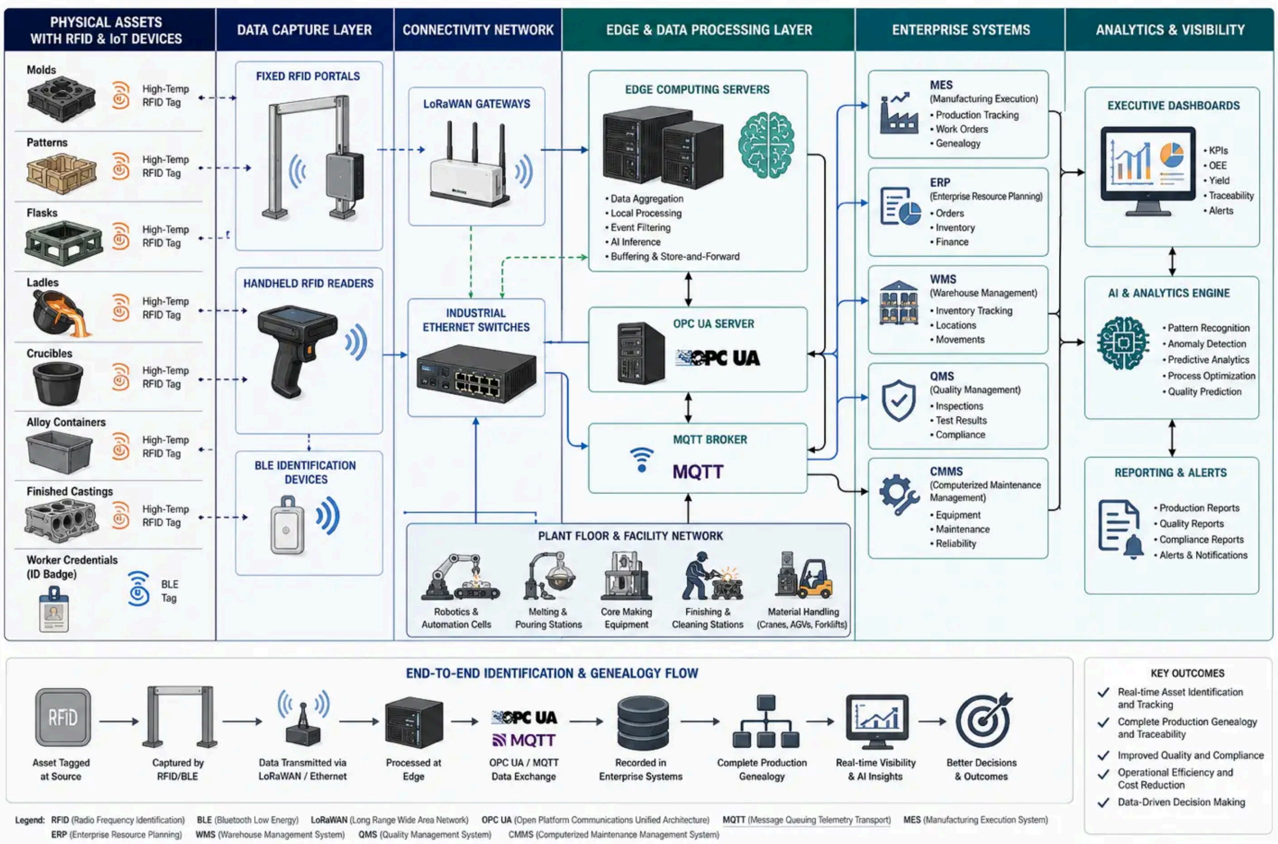
Traditional paper travelers, manual barcode recording, spreadsheets, and handwritten production logs often make it difficult to determine the real-time location of production assets, personnel, or heat lots. As production volumes increase, these limitations can affect production scheduling, inventory accuracy, tooling utilization, and quality investigations.

FoundryCast AI addresses these challenges by combining AI with industrial Internet of Things (IoT) identification technologies to create reliable digital identification throughout casting operations. AI and IoT integrates AI with RFID identification, Bluetooth Low Energy (BLE), LoRaWAN connectivity, GPS yard location technologies, edge computing, industrial communication networks, and enterprise manufacturing software. The objective is not environmental sensing but accurate identification, location awareness, and production traceability across the entire manufacturing lifecycle.

Rather than providing generic manufacturing software, FoundryCast AI is engineered specifically for metal casting environments, including iron foundries, steel foundries, aluminum foundries, investment casting facilities, permanent mold casting operations, high-pressure die casting plants, gravity die casting, shell molding, and precision casting manufacturers. The result is greater operational visibility, improved furnace workforce accountability, enhanced tooling utilization, accurate alloy inventory management, digital casting genealogy, and complete heat lot traceability from raw material receiving through final shipment.

AIoT-Enabled Foundry Enterprise Block Diagram for RFID Identification and Production Genealogy

This engineering block diagram illustrates how RFID, BLE, LoRaWAN, Industrial Ethernet, edge computing, OPC UA, and MQTT connect physical foundry assets with enterprise software platforms. It demonstrates end-to-end identification, production genealogy, enterprise data flow, and AI-driven analytics across MES, ERP, WMS, QMS, CMMS, warehouse operations, and executive dashboards.



Why AI and IoT Matters in Foundries & Casting

Metal casting operations involve thousands of reusable production assets and continuously moving production batches. Patterns, molds, flasks, ladles, crucibles, chill plates, fixtures, pallets, alloy containers, and finished castings frequently move among molding lines, furnace areas, pouring stations, cooling zones, machining departments, inspection laboratories, warehouses, and shipping docks. Manual tracking often results in production delays when assets cannot be located quickly or production records require extensive reconciliation.

AI and IoT enables automated identification throughout these workflows by associating unique digital identities with workers, production equipment, tooling, and castings. Industrial RFID supports reliable identification of metal production assets using specialized on-metal and high-temperature RFID tags engineered for harsh foundry environments. BLE wearable devices improve workforce accountability around induction furnaces, cupola furnaces, electric arc furnace support areas, pouring stations, and maintenance zones. LoRaWAN provides long-range communication across large industrial campuses, while GPS assists with locating outdoor inventory, returnable containers, and scrap storage areas.

Identification events are processed locally through edge computing devices before securely synchronizing with Manufacturing Execution Systems (MES), Enterprise Resource Planning (ERP) software, Warehouse Management Systems (WMS), Quality Management Systems (QMS), and Computerized Maintenance Management Systems (CMMS). Standard industrial communication protocols such as OPC UA, MQTT, Modbus TCP, and REST APIs support reliable interoperability with existing production software while minimizing disruption to established workflows.

The resulting digital production records help engineers, production supervisors, quality managers, maintenance personnel, warehouse operators, and supply chain teams make better-informed decisions using accurate operational information rather than manual estimates.

Operational Challenges Addressed

Modern foundries continue to increase automation while facing greater demands for quality assurance, traceability, workforce safety, inventory accuracy, and production efficiency. Identification gaps frequently occur because production assets are exposed to elevated temperatures, abrasive materials, heavy equipment movement, and complex routing between manufacturing departments.

FoundryCast AI helps address operational challenges such as:

- Restricted access to furnace, pouring, and molten metal handling areas
- Limited visibility of furnace crew locations during production shifts
- Difficulty locating molds, patterns, flasks, and reusable tooling
- Ladle and crucible utilization tracking
- Alloy inventory reconciliation
- Scrap segregation and return scrap management
- Production batch identification
- Casting work-in-progress visibility
- Heat number association throughout manufacturing
- Production genealogy documentation
- Material certification management
- Dimensional inspection record association
- NDT documentation linkage
- Warehouse inventory accuracy
- Outdoor yard inventory management
- Shipping verification
- MES and ERP data consistency

- Regulatory and customer documentation readiness

These capabilities help establish reliable digital production records that support operational excellence, quality assurance, continuous improvement initiatives, and long-term manufacturing competitiveness.

AI Functions for AIoT-Enabled Foundries & Casting Operations

Foundry operations require continuous coordination between melting, mold preparation, core making, pouring, cooling, shakeout, fettling, machining, inspection, warehousing, and shipping. Every production stage depends on accurate identification of personnel, reusable production assets, alloy materials, production batches, and finished castings. Rather than relying on manual travelers or handwritten production records, AI and IoT automates identification and location awareness using industrial RFID, BLE, LoRaWAN, GPS, edge computing, and enterprise software integration.

Unlike general manufacturing solutions, FoundryCast AI focuses specifically on identification and location technologies for harsh foundry environments where high temperatures, molten metal, metal-rich surroundings, overhead cranes, forklifts, and continuous production movement require durable identification methods and reliable communication.

Workforce & Furnace Safety System

Personnel safety is a top priority in melting, pouring, and molten metal handling operations. Furnace operators, maintenance technicians, crane operators, refractory specialists, mold technicians, and quality personnel frequently move between controlled work zones where access authorization and personnel accountability are essential.

AI and IoT improves workforce visibility through digital identification rather than manual attendance records, helping supervisors maintain awareness of authorized personnel without interrupting production.

Furnace Crew Location Analytics

Industrial BLE wearable identification devices and RFID personnel credentials help identify authorized workers assigned to induction furnaces, cupola furnaces, holding furnaces, melt decks, pouring aisles, refractory maintenance areas, and charge preparation stations.

Operational benefits include:

- Improved workforce accountability
- Faster personnel verification
- Digital shift records
- Reduced manual attendance documentation
- Improved emergency response coordination
- Historical workforce movement analysis

Melt Shop Access Analytics

Restricted production areas require controlled entry to protect personnel and maintain operational discipline. RFID-enabled credentials automate entry authorization while maintaining digital records for audits and operational reviews.

Typical controlled areas include:

- Furnace rooms
- Charge preparation
- Molten metal transfer routes
- Ladle maintenance stations
- Refractory workshops
- Metallurgical laboratories
- Maintenance workshops
- Control rooms

High-Temperature Zone Compliance

AI software evaluates identification events occurring near designated production areas to confirm that only authorized personnel enter restricted zones during active production. Historical records support internal safety reviews, operational investigations, and contractor management.

Foundry Workforce Movement Analytics

Movement analytics provide operational insight into workforce allocation across molding departments, core rooms, pouring lines, shakeout operations, heat treatment areas, machining cells, inspection laboratories, and warehouses. Historical trends assist production planners with staffing optimization and resource allocation.

Tooling, Asset & Inventory Analytics

Reusable production tooling represents a major operational investment. Patterns, core boxes, molds, flasks, chill plates, fixtures, pouring equipment, and transport containers continuously circulate throughout production. Rapid identification improves utilization while reducing unnecessary replacement, production delays, and manual searching.

Mold and Tooling Analytics

Industrial RFID enables continuous identification of:

- Patterns
- Match plates
- Core boxes
- Mold assemblies
- Flasks
- Chill plates
- Fixtures

- Production pallets
- Maintenance tooling

Production supervisors can determine current asset location, utilization history, maintenance intervals, and production availability.

Ladle and Crucible Analytics

Molten metal transfer depends on reliable availability of ladles, crucibles, transfer vessels, pouring shanks, and refractory-lined equipment. Automated identification supports utilization analysis, maintenance scheduling, refractory inspection planning, and production readiness.

Historical utilization records also assist engineering teams when evaluating equipment replacement strategies.

Alloy Inventory Forecasting

Foundries manage diverse charge materials, including:

- Pig iron
- Steel scrap
- Return scrap
- Aluminum ingots
- Ferroalloys
- Nickel additions
- Chromium additions
- Silicon alloys

- Magnesium treatments
- Inoculants
- Flux materials

RFID identification automates receiving, warehouse movement, production staging, and inventory reconciliation. AI software analyzes historical consumption to support production scheduling and purchasing decisions while reducing material shortages.

Scrap Metal Inventory Analytics

Internal return scrap, machining chips, gating systems, risers, rejected castings, and recyclable materials represent valuable production resources.

Automated identification helps improve:

- Scrap segregation
- Material classification
- Inventory reconciliation
- Storage location management
- Internal recycling workflows
- Production yield analysis

Casting Production Analytics

Casting operations involve complex production routing across multiple manufacturing departments. Identification events automatically create digital production records that improve visibility without increasing operator workload.

Casting Flow Analytics

Each production batch progresses through:

- Charge preparation

- Furnace melting
- Melt chemistry verification
- Spectrometer analysis
- Pouring
- Mold cooling
- Shakeout
- Fettling
- Shot blasting
- Heat treatment
- Machining
- Inspection
- Packaging
- Warehouse storage
- Shipping

AI software associates identification events across every production stage, providing production supervisors with comprehensive operational visibility.

Pour-to-Finish Cycle Analytics

Tracking production duration between pouring and shipment enables production managers to evaluate operational efficiency, improve scheduling accuracy, and identify bottlenecks affecting manufacturing throughput.

Mold Cycle Time Prediction

Historical utilization records allow AI software to estimate mold turnaround intervals, improving production planning and tooling availability while reducing idle production capacity.

Casting Queue Optimization

Production queues frequently develop before shakeout, machining, inspection, or packaging. AI software evaluates identification records to help supervisors balance workloads, prioritize production orders, and improve throughput without disrupting manufacturing schedules.

Heat Lot & Quality Traceability

Complete production genealogy has become increasingly important for automotive, aerospace, mining equipment, rail transportation, defense, heavy machinery, energy infrastructure, and industrial equipment manufacturers.

Digital traceability supports regulatory compliance, customer certification, warranty investigations, and quality assurance.

Heat Lot Traceability Analytics

Every heat number can remain digitally associated with:

- Charge materials
- Melt chemistry
- Alloy additions
- Furnace operations
- Pouring sequence

- Mold identification
- Production batches
- Finished castings
- Inspection records
- Customer shipments

This digital genealogy greatly simplifies product documentation throughout the manufacturing lifecycle.

Casting Defect Root Cause Analytics

When defects such as shrinkage, porosity, inclusions, cold shuts, hot tears, misruns, dimensional deviations, or surface discontinuities are identified, AI software correlates production history with identification records to simplify engineering investigations and continuous improvement activities.

Material Certification Traceability

Material certificates, chemical analysis reports, mechanical property documentation, heat treatment records, dimensional inspection reports, and non-destructive testing results can be digitally associated with each production batch, improving documentation accuracy and retrieval efficiency.

IoT Software for AIoT-Enabled Foundries & Casting Facilities

Foundries require specialized software capable of supporting production identification, operational documentation, inventory control, production genealogy, and enterprise integration.

FoundryCast AI software is modular, allowing organizations to deploy functionality according to operational priorities while integrating with existing manufacturing systems.

Workforce Identification Software

The workforce identification module supports personnel accountability throughout production while reducing manual administrative activities.

Core capabilities include:

- Wearable worker credential management
- Furnace area access control
- Shift attendance management
- Emergency muster management
- Contractor identification
- Visitor authorization
- Department assignment
- Qualification verification
- Workforce reporting
- Historical identification records

Tooling & Inventory Software

Automated identification improves utilization of reusable production assets while supporting inventory accuracy.

Software modules include:

- Pattern asset management
- Match plate management
- Core box identification
- Mold asset management
- Flask tracking
- Ladle identification
- Crucible management
- Raw material inventory collection
- Alloy inventory management
- Scrap inventory management
- Finished casting inventory software
- Warehouse pallet identification

Production Tracking Software

Production tracking software creates digital manufacturing records throughout the casting lifecycle.

Capabilities include:

- Casting work-in-progress tracking

- Heat number recording
- Pour sequence documentation
- Production batch management
- Mold association
- Production routing
- Queue visibility
- Manufacturing history
- Shipping verification

Traceability Management Software

Comprehensive production documentation simplifies quality assurance and customer reporting.

Capabilities include:

- Heat lot traceability
- Material certification management
- Production genealogy
- Chemical analysis association
- Heat treatment documentation
- Inspection record management

- NDT documentation linkage
- Digital production records
- Historical archive management

IoT Hardware Technologies for AIoT-Enabled Foundries & Casting

Harsh foundry environments demand identification hardware designed for elevated temperatures, abrasive dust, vibration, moisture, heavy equipment movement, and metallic interference. Hardware selection should prioritize durability, read reliability, environmental suitability, and compatibility with existing industrial infrastructure.

Foundry Identification Devices

Typical hardware includes:

- High-temperature RFID tags
- On-metal RFID tags
- Ceramic RFID tags
- Heat-resistant worker identification badges
- Rugged BLE wearable devices
- Mold RFID tags

- Flask RFID tags
- Pattern RFID tags
- Ladle RFID tags
- Crucible RFID tags
- Finished casting RFID labels
- Fixed RFID portals
- Industrial handheld RFID readers
- Forklift-mounted RFID readers
- Vehicle-mounted RFID readers

AI and RFID Technologies

RFID serves as the primary identification technology for most foundry assets because it enables reliable identification without requiring direct visual contact.

Typical applications include:

- Workforce identification
- Asset identification
- Tool tracking
- Mold tracking
- Heat lot tagging

- Inventory management
- Warehouse identification
- Shipping verification
- Production documentation

AI and BLE Technologies

BLE identification complements RFID by supporting personnel identification and authorized movement throughout production facilities.

Common applications include:

- Workforce identification
- Authorized access verification
- Contractor management
- Maintenance crew coordination
- Emergency mustering
- Workforce accountability

AI and Industrial Connectivity

Reliable communication infrastructure ensures identification events are securely exchanged with enterprise software.

Common technologies include:

- LoRaWAN
- Industrial Ethernet
- Industrial Wi-Fi
- Fiber optic backbone
- GPS yard identification
- OPC UA
- MQTT
- Modbus TCP
- REST APIs
- Edge computing
- Industrial middleware

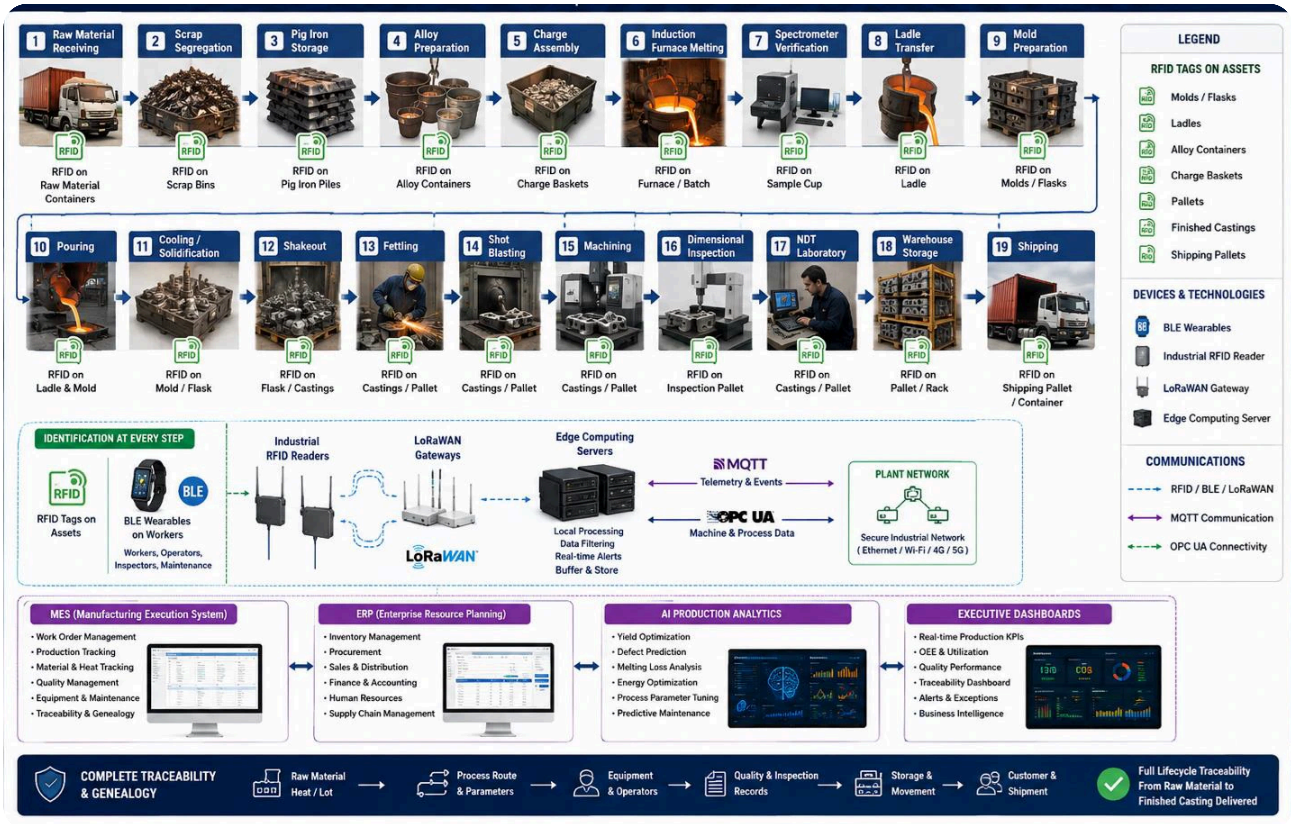
Foundries & Casting System Integration

Successful AI and IoT implementations in foundries depend on reliable integration with existing manufacturing software rather than replacing established production systems. Identification data becomes significantly more valuable when automatically associated with production orders, heat numbers, batch records, inspection documentation, inventory transactions, maintenance activities, and shipping records.

FoundryCast AI is designed to integrate with industrial software commonly used throughout the Primary Metals Industry while preserving established manufacturing workflows and minimizing operational disruption. The objective is to create a unified digital production record that follows each casting from charge preparation through customer shipment.

AIoT-Enabled Foundry Workflow for End-to-End Identification and Production Genealogy

This workflow diagram illustrates complete identification and production genealogy across the foundry manufacturing lifecycle, from raw material receiving to shipping. It highlights RFID-tagged assets, BLE worker wearables, industrial RFID readers, LoRaWAN gateways, edge computing, OPC UA and MQTT communications, and integration with MES, ERP, AI production analytics, and executive dashboards to enable real-time traceability, quality control, and operational visibility.



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Deployment Models

Every foundry has different cybersecurity policies, production requirements, and information technology strategies. FoundryCast AI supports multiple deployment models that align with corporate governance, regulatory requirements, and operational objectives.

Cloud Foundry Tracking Software

Cloud deployment supports centralized management across multiple foundries, regional manufacturing facilities, or geographically distributed operations. Authorized users can securely access production records, operational dashboards, inventory information, and traceability documentation from approved locations.

Typical advantages include:

- Centralized software administration
- Multi-site production visibility
- Enterprise reporting
- Simplified software maintenance
- Secure remote access
- Scalable historical data storage
- Centralized user management
- Faster deployment of software updates

Cloud deployment is often preferred for organizations managing multiple production facilities that require consolidated operational reporting and standardized manufacturing documentation.

On-Premises Foundry Tracking Software

Many steel, iron, aluminum, and specialty alloy foundries prefer local deployment because production systems operate within tightly controlled industrial networks. On-premises deployment provides direct ownership of infrastructure, databases, cybersecurity policies, backup strategies, and software maintenance schedules.

Typical advantages include:

- Complete local data governance
- Lower dependency on external connectivity
- High-performance local processing
- Compliance with internal cybersecurity standards
- Direct control of maintenance windows
- Simplified integration with existing production infrastructure
- Greater customization flexibility
- Low-latency communication within manufacturing facilities

Organizations handling sensitive customer programs, defense manufacturing, or proprietary metallurgical processes frequently select this deployment model.

Enterprise Connectivity

Reliable interoperability is essential for digital foundry operations. Identification events generated throughout production should automatically enrich enterprise manufacturing records without requiring duplicate operator input.

FoundryCast AI supports integration with:

- Manufacturing Execution Systems (MES)

- Enterprise Resource Planning (ERP)
- Warehouse Management Systems (WMS)
- Quality Management Systems (QMS)
- Computerized Maintenance Management Systems (CMMS)
- Laboratory Information Management Systems (LIMS)
- Production scheduling software
- Shipping and logistics software
- Product Lifecycle Management (PLM) systems
- BI reporting tools

Industrial communication commonly utilizes:

- OPC UA
- MQTT
- Modbus TCP
- REST APIs
- SQL database connectivity
- XML and JSON data exchange
- Secure HTTPS communication

This interoperability allows production records, inventory transactions, maintenance activities, inspection documentation, and shipping confirmations to remain synchronized throughout the manufacturing lifecycle.

Foundry Edge Processing Software

Edge computing improves operational responsiveness by processing identification events close to production activities before synchronizing selected information with enterprise software.

Typical edge processing functions include:

- RFID event filtering
- Duplicate read elimination
- Worker credential validation
- Production event association
- Local database caching
- Temporary communication buffering
- Production continuity during network interruptions
- Local operational reporting
- Device health monitoring
- Secure encrypted communication

Local processing reduces unnecessary network traffic while supporting continuous operation in demanding industrial environments.

Casting Systems Middleware

Industrial middleware coordinates communication among RFID readers, BLE infrastructure, enterprise software, databases, and reporting applications.

Middleware responsibilities include:

- RFID device management
- BLE infrastructure management
- Device authentication
- Communication routing
- Data normalization
- Event filtering
- API management
- Enterprise integration
- Audit logging
- User authentication
- Role-based access control
- Secure data exchange

Middleware also simplifies future expansion by allowing additional identification hardware or enterprise applications to be integrated without major software redesign.

Operational Benefits for Foundries & Casting

Accurate identification throughout casting operations provides measurable operational improvements that extend beyond production reporting. Automated identification reduces manual administrative activities while providing engineering, production, quality, maintenance, and logistics teams with more consistent operational information.

Organizations commonly achieve improvements in:

- Furnace workforce accountability
- Restricted area access management
- Personnel identification accuracy
- Mold availability
- Pattern utilization
- Flask utilization
- Core box management
- Ladle lifecycle management
- Crucible availability
- Alloy inventory accuracy
- Charge material visibility
- Scrap reconciliation
- Production batch visibility
- Casting work-in-progress tracking

- Heat lot documentation
- Production genealogy
- Inspection record association
- Warehouse inventory accuracy
- Outdoor inventory visibility
- Shipping verification
- Customer documentation readiness
- Regulatory compliance support
- Enterprise reporting consistency

These improvements contribute to higher production efficiency, better quality documentation, improved operational coordination, and reduced administrative overhead.

Engineering Best Practices for AI and IoT Deployments in Foundries

Successful implementation begins with a detailed engineering assessment of production workflows, identification objectives, environmental conditions, and enterprise software requirements. Identification technologies should complement existing manufacturing practices rather than introduce unnecessary complexity.

Recommended engineering practices include:

- Conduct a comprehensive identification workflow assessment before deployment.
- Document production routing from raw material receiving through shipment.
- Identify all reusable production assets, including patterns, molds, flasks, core boxes, ladles, crucibles, fixtures, and returnable containers.
- Select RFID tags specifically designed for high-temperature and metal-rich environments.
- Perform RFID site surveys to evaluate read performance around large metallic structures, overhead cranes, and production equipment.
- Establish standardized heat number and production batch identification procedures.
- Define workforce identification policies for furnace operators, maintenance personnel, contractors, and visitors.
- Integrate identification events with MES, ERP, WMS, QMS, and CMMS software.
- Use secure communication protocols such as OPC UA, MQTT, and encrypted HTTPS for enterprise connectivity.
- Validate identification accuracy during pilot deployments before full-scale implementation.
- Train operators, maintenance technicians, warehouse personnel, and production supervisors using standardized operating procedures.
- Periodically review system performance, identification accuracy, and workflow efficiency to support continuous improvement.

A phased deployment strategy minimizes production disruption while enabling engineering teams to refine workflows based on operational experience.

Why Choose FoundryCast AI

FoundryCast AI was developed using practical experience gained from industrial AI and IoT deployments across manufacturing environments where reliable identification, production traceability, and enterprise integration are critical to operational success. Rather than offering generic manufacturing software, FoundryCast AI focuses on the unique requirements of foundries and casting operations, including furnace workforce accountability, mold and tooling identification, alloy inventory management, production genealogy, and heat lot traceability.

Created within Aperture Venture Studio with support from GAO, FoundryCast AI builds upon more than two decades of IoT experience serving thousands of customers and successfully delivering thousands of IoT projects across industrial sectors, including the Primary Metals Industry. Extensive investment in research and development, rigorous quality assurance processes, and experienced engineering support contribute to reliable implementations and long-term operational value.

The company is led by Ph.D.-level professionals from leading universities and is supported by experienced engineers, technology specialists, and strategic partners. Over the years, the broader organization has supported numerous Fortune 500 manufacturers, internationally recognized research institutions, prestigious universities, and government agencies in the United States and Canada. This depth of experience enables FoundryCast AI to deliver technically sound, scalable solutions that align with real-world foundry operations and enterprise manufacturing requirements.

Contact FoundryCast AI

Whether your organization operates a gray iron foundry, ductile iron foundry, steel foundry, aluminum casting facility, investment casting operation, shell molding plant, permanent mold foundry, or high-pressure die casting facility, FoundryCast AI provides practical AI and IoT identification solutions tailored to the operational requirements of modern metal casting.

Our engineering specialists collaborate with manufacturing organizations to evaluate production workflows, identify appropriate identification technologies, define phased implementation strategies, and integrate AI and IoT software with existing enterprise systems. Every deployment is designed to improve operational visibility while preserving established manufacturing practices and minimizing production disruption.

We can assist with:

- AI and IoT solution planning
- Industrial RFID deployment strategy
- High-temperature RFID tag selection
- Workforce identification system design
- Furnace access control planning
- Mold and tooling identification
- Pattern and flask tracking
- Ladle and crucible management
- Alloy inventory optimization
- Scrap inventory reconciliation
- Production genealogy implementation
- Heat lot traceability
- Casting work-in-progress visibility
- MES integration

- ERP synchronization
- Warehouse identification
- Industrial edge computing
- Enterprise software integration
- Engineering consulting
- Deployment planning and optimization

Whether your objective is improving furnace workforce accountability, reducing tooling search time, strengthening production documentation, increasing inventory accuracy, or enhancing quality traceability, FoundryCast AI provides technically sound AI and IoT solutions developed specifically for foundry operations.

Advancing Digital Transformation in Foundries & Casting with AI and IoT

Foundries continue to evolve as digital manufacturing technologies improve visibility across every stage of the casting lifecycle. Accurate identification of personnel, production assets, molds, tooling, alloy materials, production batches, and finished castings has become essential for maintaining productivity, quality assurance, regulatory compliance, and customer confidence.

FoundryCast AI combines industrial RFID, BLE, LoRaWAN, GPS yard tracking, edge computing, and enterprise software integration to create reliable AI and IoT identification solutions for the unique operating conditions of the Primary Metals Industry. By supporting workforce identification, furnace access control, tooling management, alloy inventory visibility, casting work-in-progress tracking, production genealogy, and heat lot traceability, the solution enables organizations to build accurate digital production records from raw material receiving through shipment.

The result is improved operational coordination, more efficient asset utilization, stronger quality documentation, faster engineering investigations, better inventory control, and a scalable foundation for continuous improvement across modern foundries and casting facilities.

Contact Us

AIoT-based people tracking, access control, asset tracking, inventory control, work-in-progress, and traceability for the Foundries & Casting industry.

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Location

FoundryCast AI

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